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## Integrated circuit with off-state diagnosis for driver channels


### Abstract

An integrated circuit includes a plurality of power transistor driver channels for driving external loads. The driver channels can be selectively configured as high side or low side driver channels. The integrated circuit includes, for each driver channel, a respective analog test circuit and a respective controller. The integrated circuit includes a single counter connected to each of the controllers for simultaneously controlling off-state diagnosis timing windows for the driver channels.



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## Background/Summary

BACKGROUND

Technical Field

(1) The present disclosure is related to integrated circuits, and more particularly, to diagnostics for high side/low side driver channels.

Description of the Related Art

(2) Some integrated circuits have dedicated driver channels for driving loads external to the integrated circuits. Each driver channel may have a power transistor having source and drain terminals coupled to dedicated output terminals of the integrated circuit. Internal driver circuitry of the integrated circuit can drive the gate terminal of the transistor.

(3) In some cases, it is possible that a fault condition may occur at the driver channel. Such fault conditions can include open load conditions or short-circuit conditions. Accordingly, some integrated circuits have included off-state diagnosis circuitry that attempts to detect fault conditions at the driver channels prior to turning on the corresponding transistors and driving the load. However, such off-state diagnosis circuitry typically consumes a large amount of area of the integrated circuit. The result is that valuable integrated circuit area cannot be utilized for the intended function of the integrated circuit.

(4) All of the subject matter discussed in the Background section is not necessarily prior art and should not be assumed to be prior art merely as a result of its discussion in the Background section. Along these lines, any recognition of problems in the prior art discussed in the Background section or associated with such subject matter should not be treated as prior art unless expressly stated to be prior art. Instead, the discussion of any subject matter in the Background section should be treated as part of the inventor's approach to the particular problem, which, in and of itself, may also be inventive.

BRIEF SUMMARY

(5) Embodiments of the present disclosure provide an integrated circuit including a plurality of driver channels that can be selectively configured as either high side (HS) or low side (LS) channels. The integrated circuit includes off-state diagnosis circuitry that detects fault conditions at the driver channels while the driver channels are off. The off-state diagnosis circuitry includes a counter that is shared by all of the channels. The counter provides timing windows to all of the driver channels for performing off-state diagnosis. Because the counter is shared by all of the driver channels, the off-state diagnosis circuitry consumes only a small amount of integrated circuit area. Furthermore, off-state diagnosis is performed in a highly efficient and effective manner.

(6) Each driver channel includes an internal power transistor having source and drain terminals coupled to the dedicated output terminals of the integrated circuit. Each driver channel also includes a respective analog test circuit and a respective controller coupled to the analog test circuit. All of the controllers are coupled to a single counter. The counter provides the timing windows to the controllers. The controllers then selectively control the analog test circuits to perform off-state diagnosis on the driver channels in accordance with the shared timing windows.

(7) Because each driver channel can either be configured as an HS driver channel or as an LS driver channel, the counter provides timing windows for HS driver channels and timing windows for LS driver channels to all of the controllers. Each controller then selectively controls the corresponding analog test circuit in accordance with either the HS timing windows or the LS timing windows, as the case may be. The result is that off-state diagnosis can be performed for all of the HS driver channels simultaneously and for all the LS driver channels simultaneously. This ensures that off-state diagnosis is performed for all driver channels in a very short amount of time.

(8) In one embodiment, an integrated circuit includes a first driver channel, a second driver channel, and a driver control system. The first driver channel includes a first power transistor and a first output terminal. The second driver channel includes a second power transistor and a second output terminal. The driver control system includes a first analog test circuit coupled to the first

output terminal, a second analog circuit coupled to the second output terminal a first controller coupled to the first analog test circuit, a second controller coupled to the second analog test circuit, and a counter coupled to the first controller and the second controller.
(9) In one embodiment, an integrated circuit includes a plurality of driver channels for driving external loads. Each driver channel includes a drain output terminal and a source output terminal. The integrated circuit includes a plurality of analog test circuits each coupled to a respective one of the plurality of driver channels. The integrated circuit includes a counter configured to generate timing windows for each of the plurality of analog test circuits to test the respective one of the plurality of driver channels.
(10) In one embodiment, a method includes generating, with a counter of an integrated circuit, timing windows and providing the timing windows from the counter to a first controller and a second controller of the integrated circuit. The method includes controlling, with the first controller, testing of the first driver channel based on the timing windows and controlling, with the second controller, testing of the second driver channel based on the timing windows.

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## Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) Reference will now be made by way of example only to the accompanying drawings. In the drawings, identical reference numbers identify similar elements or acts. In some drawings, however, different reference numbers may be used to indicate the same or similar elements. The sizes and relative positions of elements in the drawings are not necessarily drawn to scale. For example, the shapes of various elements and angles are not necessarily drawn to scale, and some of these elements may be enlarged and positioned to improve drawing legibility.
(2) FIG. **1** is a block diagram of an integrated circuit including a plurality of driver channels and a driver control system, according to one embodiment.
(3) FIG. **2** is a block diagram of an integrated circuit, according to one embodiment.
(4) FIG. **3** is a schematic diagram of a driver control system, according to one embodiment.
(5) FIGS. **4**A and **4**B are schematic diagrams of an integrated circuit, according to one embodiment.
(6) FIG. **5**A-**5**D are graphs illustrating timing windows for performing off-state diagnosis, according to one embodiment.
(7) FIG. **6** is a flow diagram of a method for operating an integrated circuit, according to one embodiment.
DETAILED DESCRIPTION
(8) In the following description, certain specific details are set forth in order to provide a thorough understanding of various disclosed embodiments. However, one skilled in the relevant art will recognize that embodiments may be practiced without one or more of these specific details, or with other methods, components, materials, etc. In other instances, well-known systems, components, and circuitry associated with integrated circuits have not been shown or described in detail, to avoid unnecessarily obscuring descriptions of the embodiments.
(9) Unless the context requires otherwise, throughout the specification and claims which follow, the word “comprise” and variations thereof, such as, “comprises” and “comprising” are to be construed in an open, inclusive sense, that is as “including, but not limited to.” Further, the terms “first,” “second,” and similar indicators of sequence are to be construed as interchangeable unless the context clearly dictates otherwise.
(10) Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrases “in one embodiment” or

"in an embodiment" in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.
(11) As used in this specification and the appended claims, the singular forms "a," "an," and "the" include plural referents unless the content clearly dictates otherwise. It should also be noted that the term "or" is generally employed in its broadest sense, that is as meaning "and/or" unless the content clearly dictates otherwise.
(12) FIG. **1** is a block diagram of an integrated circuit **100**, according to one embodiment. The integrated circuit **100** includes a plurality of driver channels **102** and a driver control system **104**. As will be set forth in more detail below, the driver control system **104** is configured to perform off-state diagnosis on the driver channels **102** in an efficient and effective manner.
(13) Each driver channel **102** includes a respective power transistor and a pair of output terminals **108**. The pair of output terminals **108** of the driver channel correspond to output terminals of the integrated circuit **100**. The output terminals **108** enable the power transistor **106** to drive a current through a load external to the integrated circuit.
(14) In one embodiment, for each driver channel **102**, the power transistor **106** includes a source terminal coupled to one output terminal **108** of the pair of output terminals **108** and a drain terminal coupled to the other of the pair of output terminals **108**. Each driver channel **102** can be selectively configured as either an HS driver or an LS driver. In the case of an LS driver, an external load is coupled between an external high voltage source and the drain terminal of the power transistor **106**. The source terminal is coupled to ground. In the case of an HS driver, the drain terminal of the power transistor **106** is coupled to an external high voltage source and the external load is coupled between the source terminal and ground.
(15) The driver control system **104** controls the driver channels **102**. In particular, the control system receives **104** input data indicating, for each driver channel **102**, whether the driver channel **102** is configured as an HS driver channel, an LS driver channel, or is not connected for operation. The driver control system **104** drives the gate terminals of the power transistors **106** in accordance with the corresponding connection scheme to drive currents through the external loads.
(16) The driver control system **104** also performs off-state diagnosis procedures on each of the driver channels **102** prior to driving the gate terminals of the power transistors **106**. The off-state diagnosis detects whether or not there are faults present at the driver channels **102**. The faults can include short circuit faults and open load faults. The driver control system **104** can detect other types of faults without departing from the scope of the present disclosure.
(17) The driver control system **104** includes, for each driver channel **102**, a respective gate driver **109**. The gate drivers **109** each selectively drive the gate terminal of the corresponding power transistor **106**. Accordingly, the gate drivers **109** apply voltages to the gate terminals of the power transistors **106**.
(18) The driver control system **104** includes, for each driver channel **102**, a respective analog test circuit **110**. Each analog test circuit **110** performs off-state diagnosis procedures for the corresponding driver channel **102**. Each analog test circuit **110** is coupled to the pair of output terminals **108** of the respective driver channel **102**. The analog test circuits **110** can drive one or more tests currents and can then detect the voltages that appear at the output terminals **108** in order to determine whether or not the fault is present. In some cases, driving a first test current determines whether there is a fault present. If the fault is present after driving the first current, then the analog test circuit **110** drives a second current. The voltage present at the output terminals **108** while driving the second current determines which type of fault is present. Further details regarding the analog test circuits **110** and regarding the timing of the analog test circuits **110** will be provided below. Various schemes and circuit structures can be utilized for the analog test circuits **110** without departing from the scope of the present disclosure.
(19) The driver control system **104** includes, for each driver channel **102**, a respective controller

**112**. Each controller **112** is coupled to the analog test circuit **110** associated with the respective driver channel **102**. The controller **112** can control the analog test circuit **110**. Each controller **110** stores information indicating whether the corresponding driver channel is configured as an LS driver, an HS driver, or is not coupled for operation. If the corresponding driver channel **102** is not coupled for operation, then the controller **110** may not cause the analog test circuit **110** to perform off-state diagnosis procedures. If the corresponding driver channel **102** is configured as an HS driver channel, then the controller **112** controls the analog test circuit **110** to perform HS off-state diagnosis procedures. If the corresponding driver channel **102** is configured as an LS driver channel, then the controller **112** controls the analog test circuit **110** to perform LS off-state diagnosis procedures.

(20) Each controller **112** can interpret the signals received from the analog test circuit **110**. The controller **112** can determine whether or not a fault is present, and what type of fault is present, based on the signals provided by the analog test circuit **110**. The timing associated with performing off-state diagnosis procedures will be described in more detail below.

(21) Each controller **112** is coupled to the corresponding gate driver **109** of the respective driver channel **102**. The controller **112** controls the gate driver **109** to drive the gate terminal of the corresponding power transistor **106**. If the off-state diagnosis indicates that there is a fault present, then the fault information is recorded in a register.

(22) The driver control system **104** includes a counter **114**. The counter **114** is coupled to each of the controllers **112**. The counter **114** provides timing windows to the controllers **112** for performing off-state diagnosis. The counter **114** provides timing windows for both HS and LS off-state diagnosis to each of the controllers **112**. As each controller **112** is coupled to a driver channel **102** configured as either an LS channel or an HS channel, each controller **112** coupled to an LS channel can ignore the HS timing windows, and each controller **112** coupled to an HS channel can ignore the LS timing windows.

(23) As described herein, a timing window may correspond to a duration for which a particular test current is driven by an analog test circuit **110**. In one example, there are two HS test currents and two LS tests currents. Accordingly, the HS timing windows include a first duration for the first HS test current and a second duration for the second HS test current. The LS timing windows include a first duration for the first LS timing window and a second duration for the second LS timing window.

(24) Because the counter **114** provides all of the timing windows to each of the controllers **112** simultaneously, very little integrated circuit area is consumed for circuitry directed to generating timing windows for the various driver channels **102**. Additionally, because the timing windows are provided from a single counter **114** to all the controllers **112** simultaneously, the off-state diagnosis for each driver channel **102** can be performed substantially simultaneously. In one embodiment, the off-state diagnosis for the HS driver channels is performed simultaneously before or after the simultaneous off-state diagnosis for LS drivers.

(25) FIG. **2** is a block diagram of an integrated circuit **100**, according to one embodiment. The FIG. **2**, the integrated circuit **100** includes five driver channels **102***a*-**102***e*. Each driver channel **102***a-e* includes a respective power transistor **106***a-e*. Each driver channel **102***a-e* also includes a drain output terminal D**1**-D**5** and a source output terminal S**1**-S**5**. The drain output terminals D**1**-D**5** are each coupled to a drain terminal of the corresponding power transistor **106***a-e*. The source output terminals S**1**-S**5** are each coupled to the source terminal of the corresponding power transistor **106***a-e*.

(26) Each of the driver channels **102***a-e* can be configured as either an HS or an LS driver channel. If a driver channel is configured as an HS driver channel, then an external load will be coupled between the source output terminal and ground and the drain output terminal will be coupled to an external high voltage source, such as a high-voltage battery. If a driver channel is configured as an LS channel, then an external load will be coupled between the external high voltage source and the

drain terminal and the source terminal will be coupled to ground. In the examples given herein, each of the power transistors **106***a-e* are N-type power MOSFETs. Alternatively, one or more of the power transistors can include P-type power MOSFETs. The power transistors can include other types of transistors than MOSFETs without departing from the scope of the present disclosure.

(27) The integrated circuit **100** includes a driver control system **104**. The driver control system **104** includes, for each driver channel **102***a-e*, a respective analog test circuit **110***a-e*. The driver control system **104** includes, for each driver channel **102***a-e*, a respective controller **112***a-e* coupled to the corresponding analog test circuit **110***a-e*. The driver control system **104** also includes a counter **114**. The counter **114** is coupled to each of the controllers **112***a-e*. The driver control system **104** also includes gate drivers (not shown) each configured to drive a date of the respective power transistor **106***a-e*. The driver control system **104** of FIG. **2** can function substantially as described in relation to FIG. **1**.

(28) The integrated circuit **100** includes a plurality of terminals **131** coupled to the driver control system **104**. In the example FIG. **2** there are three terminals **131** coupled to the driver control system **104**. However, the integrated circuit **100** may include a different number of terminals **131** than are shown in FIG. **2**. The terminals **131** may include idle command terminals, input terminals, output terminals, or other types of terminals.

(29) The integrated circuit **100** includes a power supply module **120** and a plurality of terminals **137** coupled to the power supply **120**. The terminals **137** may receive a supply voltage VDD, a ground voltage, and the external high supply voltage. The external high supply voltage is also the voltage utilize externally in driving the loads for each of the driver channels **102***a-c*. The power supply **120** applies the various voltages to appropriate circuits within the integrated circuit **100**. There may be a different number of terminals **137** than shown in FIG. **2** without departing from the scope of the present disclosure.

(30) The integrated circuit **100** includes a serial peripheral interface (SPI) block **122**. The integrated circuit **100** includes a plurality of terminals **135** coupled to the SPI block **122**. The terminals **135** can include data input terminals, data output terminals, clock terminals, or other types of terminals. The integrated circuit **100** can include a different number of terminal than are shown in FIG. **2** without departing from the scope of the present disclosure. The SPI block can store data related to the fault state and the type of fault state of each driver channel. The SPI block **122** can output data, signals, or alerts related to the state of each driver channel.

(31) FIG. **3** is a block diagram of a driver control system **104**, according to one embodiment. The driver control system **104** is one example of a driver control system **104** of FIG. **1** or **2**. The driver control system **104** includes a plurality of analog test circuits **110**. Each analog test circuit **110** includes a first current source **134** and a second current source **136**. The first current source **134** is coupled between VDD and an output terminal **108** of the corresponding driver channel. The second current source **136** is coupled between the output terminal **108** and ground. The first and second current sources **134** and **136** can be selectively activated and deactivated, as will be described in further detail below. The analog test circuits **110** each include a comparator **138**. The comparator **138** includes a first input terminal coupled to the output terminal **108** and, correspondingly, to the node between the first and second current sources **134** and **136**. A second input terminal of the comparator **138** receives a threshold voltage Vth. Operation of the analog test circuit **110** will be described further below.

(32) The driver control system **104** includes a plurality of controllers **112** each coupled to a respective analog test circuit **110**. Each controller **112** includes a first relay **128** and a second relay **130**. The first relay **128** receives timing windows for the first test phases for both HS and LS off-state diagnostics. The second relay **130** receives timing windows for the second test phases for both HS and LS off-state diagnostics. The relays **128** and **130** receive control signals that determine whether they will pass the HS or LS timing windows.

(33) Each controller **112** includes a respective control logic **132**. The control logic **132** may also be

termed a control circuit. The control logic **132** receives the timing windows from the relays **128** and **130**. The control logic **132** stores data indicating whether the corresponding driver channel **102** (not shown in FIG. **3**) is configured as an HS driver channel, and LS driver channel, or not currently configured for operation. The control logic **132** provides the control signals to the relays **128** and **130** that determine whether HS or LS timing windows will be provided to the control logic **132** based on the current configuration of the corresponding driver channel **102**. The control circuit **132** provides control signals to the current sources **134** and **136** in accordance with the timing windows, as will be explained in greater detail below. The control circuit **132** also receives the output of the comparator **138**. The control circuit **132** determines whether a fault state is present, and what kind of fault state is present, based on the output of the comparator **138**, as will be described in greater detail below.

(34) The driver control system **104** also includes a counter **114**. The counter **114** includes a timer **124** and timing blocks **126**. The timer **124** outputs timing signals, such as a current clock cycle count, to the timing blocks **126**. One of the timing blocks **126** provides to the relay **128** a timing window signal T**1**HS. The timing window signal T**1**HS indicates the timing and duration for a first phase of HS off-state diagnostics. One of the timing blocks **126** provides to the relay **128** a timing window signal T**1**LS. The timing window signal T**1**LS indicates the timing and duration for a first phase of LS off-state diagnostics. One of the timing blocks **126** provides to the relay **130** a timing window signal T**2**HS. The timing window signal T**2**HS indicates the timing and duration for the second phase of HS off-state diagnostics. One of the timing blocks **126** provides to the relay **130** a timing window signal T**2**LS. The timing window signal T**2**LS indicates the timing and duration of the second phase of LS off-state diagnostics. The timer **124** may be controlled by the control logic **132**.

(35) The timer **124** may output a signal indicating a current number of clock cycles that has been counted. Each of the timing blocks **126** may be configured to begin outputting a timing beginning at a particular number of clock cycles and ending at a particular number of clock cycles. The beginning and ending number of clock cycles determine the start time and the total duration of timing windows for the various HS and LS testing phases.

(36) In one embodiment, the current source **134** is a pull-up current source and the current source **136** is a pull-down current source. When activated, the current source **134** drives a current from the supply voltage VDD. When activated, the current source **136** drives a current to ground.

(37) For a LS driver channel, the control logic **132** causes the relay **128** to pass the timing window signal T**1**LS to the control logic **132**. The control logic **132** causes the relay **130** to pass the timing window signal T**2**LS. In one embodiment, in an LS off-state diagnosis process, the control logic **132** causes the pull-down current source **136** to activate during the timing window T**1**LS. After T**1**LS is finished, the control logic **132** turns off the pull-down current source **136** and turns on the pull-up current source **134** for the duration of the timing window T**2**LS. As will be set forth in more detail below, the control logic **132** determines whether a fault is presence based on the output of the comparator **138** during or at the end of the timing window T**1**LS. The control logic **132** determines what type of fault is present based on the output of the comparator **138** during or at the end of the timing window T**2**LS. Various other schemes for determining whether there is a fault and what type of fault is present can be utilized without departing from the scope of the present disclosure.

(38) For an HS driver channel, the control logic **132** causes the relay **128** to pass the timing window signal T**1**HS to the control logic **132**. The control logic **132** causes the relay **130** to pass the timing window signal T**2**HS to the control logic **132**. In one embodiment, in an HS off-state diagnosis process, the control logic **132** causes the pull-up current source **134** to activate during the timing window T**1**HS. After T**1**HS is finished, the control logic **132** turns off the pull-up current source **134** and turns on the pull-down current source **136** for the duration of the timing window T**2**HS. As will be set forth in more detail below, the control logic **132** determines whether fault is present based on the output of the comparator **138** during or at the end of the timing window T**1**HS.

The control logic **132** determines what type of fault is present based on the output of the comparator **138** during or at the end of the timing window T**2**HS. Various other schemes for determining whether there is a fault and what type of fault is present can be utilized without departing from the scope of the present disclosure.

(39) FIG. **4**A illustrates a portion of an integrated circuit **100**, according to one embodiment. FIG. **4**A illustrates two driver channels **102***a* and **102***b*. In FIG. **4**A, the two driver channels **102***a* and **102***b* are substantially identical. Accordingly, only description of the driver channel **102***a* will be provided in relation to FIG. **4**A.

(40) The driver channel **102***a* includes a power transistor **106***a*. The power transistor **106***a* has a drain terminal coupled to the drain output terminal D**1** of the integrated circuit **100** and a source terminal coupled to the source output terminal S**1** of the integrated circuit **100**. The driver channel **102***a* includes the analog test circuit **110***a*. The analog test circuit **110***a* is substantially similar to the analog test circuits **110** described in relation to FIG. **3**. However, FIG. **4**A illustrates one embodiment of how the analog test circuit **110***a* can be selectively connected to perform off-state diagnosis on either an LS connection or an HS connection. The driver channel **102***a* includes switches Sla and S**2***a*. The logic circuit **132** selectively closes the switch Sla and opens the switch S**2***a* for off-state diagnosis when the driver channel **102***a* is configured as an LS driver channel. The logic circuit **132** selectively opens the switch Sla and closes the switch S**2***a* for off-state diagnosis when the driver channel **102***a* is configured as an HS driver channel.

(41) FIG. **4**B illustrates the integrated circuit **100** of FIG. **4**A with the driver channel **128** configured as an LS driver channel and the driver channel **102***b* configured as an HS driver channel. For the driver channel **102***a* configured as an LS driver channel, an external load L**1** is connected between a high voltage battery Vb and the drain output terminal D**1**. The source output terminal S**1** is connected to ground. The switch Sla is closed and the switch S**2***a* is open in preparation for performing off-state diagnosis.

(42) For the driver channel **102***b* configured as an HS driver channel, an external load L**2** is connected between the source output terminal S**2** and ground. The drain output terminal D**2** is coupled to the high voltage battery Vb. The switch S**1***b* is open in the switch S**2***b* is closed in preparation for performing off-state diagnosis.

(43) A description of off-state diagnosis for the LS configured channel **102***a* and the HS configured channel **102***b* will be described with reference to FIGS. **5**A-**5**D. FIG. **5**A illustrates activation of the pull-up and pulldown current sources **134***a*,*b* and **136***a*,*b* in conjunction with the timing windows T**1**LS, T**1**HS, T**2**LS and T**2**HS. The control signal LSD controls the pulldown current source **136***a* and is high during the timing window T**1**LS. The control signal LSU controls the pull-up current source **134***a* and is high during the timing window T**2**LS. The control signal HSD controls the pulldown current source **136***b* and is high during the timing window T**2**HS. The control signal HSU controls the pull-up current source **134***b* and is high during the timing window T**1**HS.

(44) At time T**1**, the timing window T**1**LS causes the LS pulldown control signal LSD to go high for a first testing phase of the LS configured channel **102***a*. This causes the pulldown current source **136***a* to turn on for the LS first testing phase.

(45) At time T**1** the timing window T**1**HS causes the HS pull-up control signal HSU to go high for a first testing phase of the HS configured channel **102***b*. This causes the pull-up current source **134***b* to turn on for the HS first testing phase.

(46) At time T**2**, the first HS testing phase ends and the current source **134***b* is turned off. At time T**2**, the second HS testing phase begins and the HS pulldown signal HSD goes high, causing the current source **134***b* to turn on.

(47) At time T**3**, the first LS testing phase ends and the current source **136***a* is turned off. At time T**3**, the second LS testing phase begins and the LS pull-up signal LSU goes high, causing the pull-up current source **134***a* to turn on.

(48) As can be seen in FIG. **5**A, the pulldown current sources are activated for a longer duration

than are the pull-up current sources during off-state diagnosis. In one example, the timing windows T**1**LS and T**2**HS the determine the control signals LSD and HSD, respectively, have a duration between 70 and 80 clock cycles counted by the counter **114**. In one example, the timing windows T**2**LS and T**1**HS have a duration between 15 and 25 clock cycles counted by the counter **114**. In one example, the timing windows T**1**LS and T**2**HS the determine the control signals LSD and HSD, respectively, have a duration between 0.8 ms and 1.2 ms. In one example, the timing windows T**2**LS and T**1**HS have a duration between 0.1 ms and 0.5 ms. Other durations can be utilized without departing from the scope of the present disclosure.
(49) FIG. **5**B illustrates a test case in which both the driver channels **102***a* and **102***b* are in an open load fault state. In FIG. **5**B, VCLS corresponds to the output of the comparator **138***a*. Prior to describing specific case of FIG. **5**B, it is beneficial to describe the signals VCHS and VCLS and what they indicate. VCHS corresponds to the output of the comparator **138***b*. If VCHS goes high during or by the end of the first HS testing phase between T**1** and T**2**, this indicates that the fault is present at the HS configured channel **102***b*. If VCHS stays low throughout the duration of the first HS testing phase between times T**1** and T**2**, this indicates that the load L**1** is present and no fault condition exists. Likewise, if VCLS goes high during or by the end of the first LS testing phase between times T**1** and T**3**, this indicates that a fault is present at the LS configured channel **102***a*. If VCLS stays low throughout the duration of the first LS testing phase between T**1** and T**3**, this indicates that the load L**1** is present and no fault condition exists.
(50) While VCLS going high during the first LS testing phase indicates the presence of a fault, this does not indicate what type of fault is present. The type of fault can be determined by the output of the comparator **138***a* (signal VCLS) during the second LS testing phase between times T**3** and T**4** when the pull-up current source **134***a* is turned on. If the voltage at the drain output terminal D**1** remains higher than the threshold voltage Vth while the pull-up current source **134***a* is active, VCLS remains high and this indicates that the fault is a short to ground fault. If the voltage at the drain output terminal D**1** goes below the threshold voltage Vth when the pull-up current source **134***a* is active, then VCLS goes low in this indicates that the fault is an open load fault.
(51) While VCHS going high during the first HS testing phase indicates the presence of a fault, this does not indicate what type of fault is present. The type of fault can be determined by the output of the comparator **138***b* (VCHS) during the second HS testing phase between times T**2** and T**4** when the pulldown current source **136***b* is turned on. If the voltage at the source output terminal S**2** remains higher than the threshold voltage Vth while the pulldown current source **136***b* is active, the CLS remains high in this indicates that the fault is a short battery. If the voltage at the source output terminal S**2** goes below the threshold voltage Vth each when the pulldown current source **134***b* is active, then VCHS goes low and this indicates that the fault is an open load fault.
(52) In FIG. **5**B, VCLS has gone high during the first LS testing phase between T**1** and T**3**. This means that while the current source **136***a* was turned on, the voltage at the drain terminal D**1** went higher than the threshold voltage Vth, indicating the presence of a fault. During the second testing phase between times T**3** and T**4**, VCLS went low, indicating that the fault is an open load fault.
(53) In FIG. **5**B, VCHS has gone high during the first HS testing phase between times T**1** and T**2**. This means that while the current source **134***b* was turned on, the voltage at the source terminal S**2** went higher than the threshold voltage Vth, indicating the presence of a fault. During the second testing phase between times T**2** and T**4**, VCHS went low, indicating that the fault is an open load fault.
(54) In FIG. **5**C, VCLS has gone high during the first LS testing phase between T**1** and T**3**. This means that while the current source **136***a* was turned on, the voltage at the drain terminal D**1** went higher than the threshold voltage Vth, indicating the presence of a fault. During the second testing phase between times T**3** and T**4**, VCLS remained high, indicating that the fault is a short to ground fault.
(55) In FIG. **5**C, VCHS has gone high during the first HS testing phase between times T**1** and T**2**.

This means that while the current source **134***b* was turned on, the voltage at the source terminal S**2** went higher than the threshold voltage Vth indicating the presence of a fault. During the second testing phase between times T**2** and T**4**, VCHS remained high, indicating that the fault is a short to battery.
(56) In FIG. **5**D, both VCLS and VCHS remained low during the first HS and LS testing phases. This indicates that no fault was present on either channel. Because the control logic **132** determines that no fault is present on either channel, the second LS and HS testing phases are not performed.
(57) In one embodiment, the control logic **132** records whether fault is present and what type of fault is present based on the results of the off-state diagnosis tests. If a fault is present, the control logic **132** can output an indication to systems external to the integrated circuit **100** that a fault is present. Furthermore, the control logic **132** disables any driver channel at which a fault is present.
(58) In FIGS. **5**A and **5**D, the HS and LS testing was performed simultaneously. However, in some embodiments, the HS and LS testing can be performed simultaneously. Both HS testing phases can be performed for all HS configured driver channels and then both LS testing phases can be performed for all LS configured driver channels. Alternatively, the LS testing phases can be performed before the HS testing phases.
(59) FIG. **6** is a flow diagram of a method **600** for performing off-state diagnosis on driver channels of the integrated circuit, according to one embodiment. The method **600** can utilize components, systems, and processes described in relation to FIGS. **1**-**5**D. At **602**, the method **600** includes generating, with a counter of an integrated circuit, timing windows. At **604**, the method **600** includes providing the timing windows from the counter to a first controller and a second controller of the integrated circuit. At **606**, the method **600** includes controlling, with the first controller, testing of the first driver channel based on the timing windows. At **608**, the method **600** includes controlling, with the second controller, testing of the second driver channel based on the timing windows.
(60) In one embodiment, an integrated circuit includes a first driver channel, a second driver channel, and a driver control system. The first driver channel includes a first power transistor and a first output terminal. The second driver channel includes a second power transistor and a second output terminal. The driver control system includes a first analog test circuit coupled to the first output terminal, a second analog circuit coupled to the second output terminal a first controller coupled to the first analog test circuit, a second controller coupled to the second analog test circuit, and a counter coupled to the first controller and the second controller.
(61) In one embodiment, an integrated circuit includes a plurality of driver channels for driving external loads. Each driver channel includes a drain output terminal and a source output terminal. The integrated circuit includes a plurality of analog test circuits each coupled to a respective one of the plurality of driver channels. The integrated circuit includes a counter configured to generate timing windows for each of the plurality of analog test circuits to test the respective one of the plurality of driver channels.
(62) In one embodiment, a method includes generating, with a counter of an integrated circuit, timing windows and providing the timing windows from the counter to a first controller and a second controller of the integrated circuit. The method includes controlling, with the first controller, testing of the first driver channel based on the timing windows and controlling, with the second controller, testing of the second driver channel based on the timing windows.
(63) The various embodiments described above can be combined to provide further embodiments. These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

## Claims

1. An integrated circuit, comprising: a first driver channel including: a first power transistor; and a first output terminal; a second driver channel including: a second power transistor; and a second output terminal; and a driver control system including: a first analog test circuit coupled to the first output terminal; a second analog test circuit coupled to the second output terminal; a first controller coupled to the first analog test circuit; a second controller coupled to the second analog test circuit; and a counter coupled to the first controller and the second controller, the counter configured to generate timing windows for off-state diagnosis for the first and second driver channels, wherein the timing windows generated by the counter comprise at least a first off-state test phase and a second off-state test phase for each driver channel, and wherein in the first off-state test phase, the respective analog test circuit detects whether a fault is present in the corresponding driver channel, and in the second off-state test phase, the respective analog test circuit identifies a type of fault when the fault is detected.
2. The integrated circuit of claim 1, wherein the counter provides, to the first controller and the second controller, a plurality of timing signals, wherein the first controller is configured to control the first analog test circuit to perform off-state diagnosis for the first driver channel in accordance with the timing signals, wherein the second controller is configured to control the second analog test circuit to perform off-state diagnosis for the first driver channel in accordance with the timing signals.
3. The integrated circuit of claim 2, wherein the first analog test circuit includes a first comparator having an input coupled to the first output terminal and an output coupled to the first controller, wherein the second analog test circuit includes a second comparator having an input coupled to the second output terminal and an output coupled to the second controller.
4. The integrated circuit of claim 3, wherein the timing signals include timing windows for both low-side driver configurations and for high-side driver configurations.
5. The integrated circuit of claim 3, wherein the first comparator is configured to output fault signals to the first controller indicating whether or not a fault is present at the first driver channel.
6. The integrated circuit of claim 4, wherein the first analog test circuit includes: a first current source coupled between a high supply voltage and the first output terminal; and a second current source coupled between the first output terminal and ground, wherein the first controller selectively activates and deactivates the first and second current source during off-state diagnosis.
7. An integrated circuit, comprising: a plurality of driver channels for driving external loads, each driver channel including: a drain output terminal; and a source output terminal; a plurality of analog test circuits each coupled to a respective one of the plurality of driver channels; and a counter configured to generate timing windows for each of the plurality of analog test circuits to perform off-state diagnosis of the respective one of the plurality of driver channels, wherein the timing windows generated by the counter comprise at least a first off-state test phase and a second off-state test phase for each driver channel, and wherein during the first off-state test phase, the respective analog test circuit detects whether a fault is present at its corresponding driver channel, and during the second off-state test phase, the respective analog test circuit determines a type of the fault upon detection.
8. The integrated circuit of claim 7, comprising a plurality of controllers each coupled to the counter and to a respective analog test circuit and configured to control the analog test circuit to perform testing operations based on the timing windows generated by the counter.
9. The integrated circuit of claim 8, wherein each driver channel can be selectively configured as a high side driver channel or a low side driver channel, wherein each controller is configured to selectively cause the corresponding analog test circuit to perform a low side testing operation or a high side testing operation depending on whether the corresponding driver channel is configured as

a high side driver channel or a low side driver channel.
10. The integrated circuit of claim 9, wherein each analog test circuit includes a comparator, wherein the corresponding controller is configured to couple an input terminal of the comparator to the drain output terminal of the corresponding driver channel when the driver channel is configured as a low side driver, and to couple the input terminal of the comparator to the source output terminal of the corresponding driver channel when the driver channel is configured as a high side driver.
11. The integrated circuit of claim 9, wherein the timing windows include high side timing windows and low side timing windows, wherein each controller is configured control the corresponding analog test signal based on the high side timing windows if the corresponding driver channel is configured as a high side driver channel and to control the corresponding analog test signal based on the low side timing windows if the corresponding driver channel is configured as a low side driver channel.
12. The integrated circuit of claim 11, wherein the high side timing windows include high side pull-up windows and high side pull-down windows, wherein the low side timing windows include low side pull-up windows and low side pull-down windows.
13. The integrated circuit of claim 12, wherein each analog test circuit is configured to identify whether or not there is a fault present at the corresponding driver channel, and, if a fault is present, to identify whether the fault is an open load fault or short circuit fault.
14. The integrated circuit of claim 13, wherein each driver channel includes a respective power transistor having: a drain terminal coupled to the drain output terminal; and a source terminal coupled to the source output terminal.
15. A method, comprising: generating, with a counter of an integrated circuit, timing windows for off-state diagnosis; providing the timing windows from the counter to a first controller and a second controller of the integrated circuit; controlling, with the first controller, testing of a first driver channel of the integrated circuit based on the timing windows; and controlling, with the second controller, testing of a second driver channel of the integrated circuit based on the timing windows, wherein the timing windows include at least a first off-state test phase and a second off-state test phase for each driver channel, wherein during the first off-state test phase, the respective driver channel is tested to detect whether a fault is present, and during the second off-state test phase, the respective driver channel is tested to determine a type of fault when a fault is detected.
16. The method of claim 15, wherein the timing windows include first timing windows and second timing windows, the method comprising: controlling, with the first controller, testing of the first driver channel based on the first timing windows; and controlling, with the second controller, testing of the second driver channel based on the second timing windows.
17. The method of claim 15, wherein the integrated circuit includes: a first analog test circuit coupled to the first controller and to the first driver channel; and a second analog test circuit coupled to the second controller and to the second driver channel.
18. The method of claim of claim 17, wherein controlling testing of first driver channel includes controlling the first analog test circuit coupled to the first driver channel.
19. The method of claim 18, comprising: receiving, for the first driver channel, data indicating whether the first driver channel is currently configured as a high side driver channel or a low side driver channel; and controlling, with the first controller, testing of the first driver channel based on the data.
20. The method of claim 19, wherein generating the timing windows includes generating high side timing windows and low side timing windows.